XCT Defect Detection

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Introduction / Background

X-Ray Computed Tomography (XCT) is used to find pores, internal holes that weaken manufactured parts.
Each scan contains 2 regions:

1. The sample - the actual material/part being scanned (a ring-shaped cross-section)
2. The pores - small voids or air pockers hidden inside that material

Pore detection is very difficult. Manual inspection is too slow to scale across high-volume productions and classical rule-based computer vision methods struggle to separate true pores from the surrounding grain texture.

This raises the central question of this project, inspired by *Woo et al.'s 2018 ECCV paper CBAM: Convolutional Block Attention Module: Can attention-based deep learning detect microscopic pores in XCT scans where classical computer vision fundamentally cannot?*
To investigate whether attention-based deep learning can succeed we applied **CBAM**, a lightweight attention module that can be inserted after each convolutional block in a CNN.

CBAM sequentially refines feature maps in two stages:

- **Channel attention** - uses average & max pooling passed through a shared MLP and a sigmoid to decide which features matter
- **Spatial attention** - which pools across channels and applies a 7x7 convolution with a sigmoid to decide where those features matter

As CBAM learns where and what to focus on, it's a `well-suited to the subtle, localized signals that distinguish a real pore from background texture.

Goal / Motivation

The goal of this project is to re-implement CBAM and integrate it into a ResNet-34 + FPN segmentation network, then test where attention improves localization in a challenging new domain: small-data industrial XCT segmentation. Working with only 14 XCT images, we compare two approaches:

- **Method 1 (Rule based):** A classical algorithm that flags anything darker than a fixed intensity threshold as a likely pore
- **Method 2 (Deep Learing with CBAM):** A network that learns from examples what pores look like

Results

Finding 1: Deep learning is necessary, not just better, for pore detection
The simple rule-based method correctly identified pores 22% of the time. The CBAM model identified them 84 % of the time. That's not "a little better"- that's a 3.8x improvement.
The rule-based method performed horribly that you couldn't use it in a factory and the CBAM was so good that you could.
This was found that pores look almost identical to normal grainy texture in the metal. Checking if spot X is darker can't tell them apart. The CBAM model learned pores have a certain shape, they cluster in certain ways, they show up at certain depths, which is exactly the kind of judgement a rule-based program can never make.

Finding 2: Both precision and recall exceed 0.90 simultaneously on pores
Precision: CBAM scores 0.92 (right 92% of the time)
Recall: CBAM scores 0.91 (catches 91% of them)
Rule-based was only able to get one of these high at a time

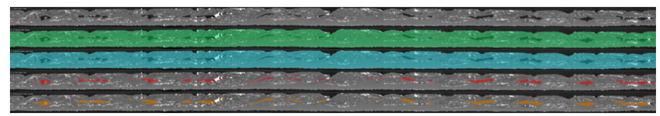
Finding 3: AI mostly helps with pores
The metal part is a clear ring shape. Both methods find it will but the rule-based method finds it 89% of the time and CBAM gets is 96% of the time. Using deep learing is for the pores.

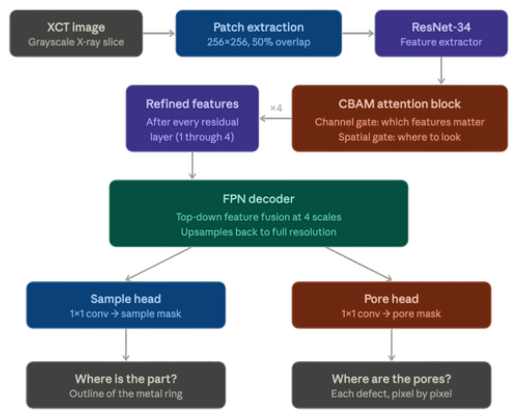
Finding 4: Image 413 failed for both methods
Image 413 had terrible scores for both methods. When two completely different methods fail on the same image in the same way, the problem isn't the methods, the problem is the image itself. Either it was labeled incorrecly by a human or the X-ray scan wasn't working properly.

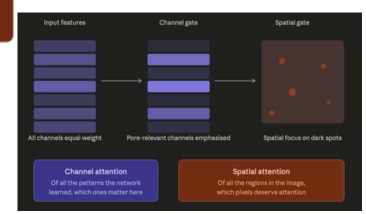
Finding 5: Getting images from only 14 images
Most deep learning proejcts use thousands of images. As a team we had strong results from 14, by cutting each image into about 33 small overlapping patches, yielding 1,306 training examples. But if only 2 images for testing, the numbers are statistically fragile. If image 413 had been swapped for a different test image, the headline number could shift noticeable which is why we used K-fold cross-validation.

Methodology

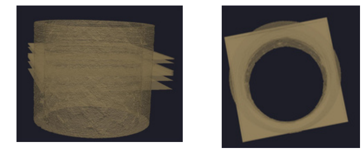
Sample - 438 | Pore - 438 | Sample - 413 | Pore - 413



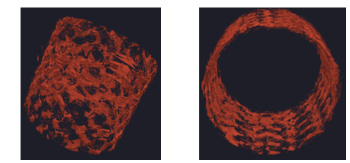




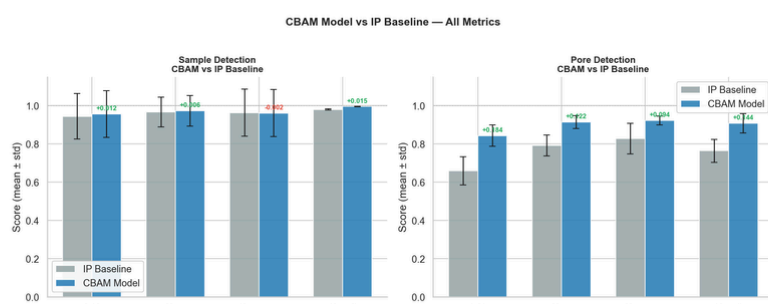
Samples - 3D



Pores - 3D



Summary Metrics



Conclusions

Attention earns its keep on the hard tasks

- Sample IoU gain: +0.063 pore IoU gain: +0.620. CBAM helps where context matters, not where geometry suffices.

Production-quality pore detection

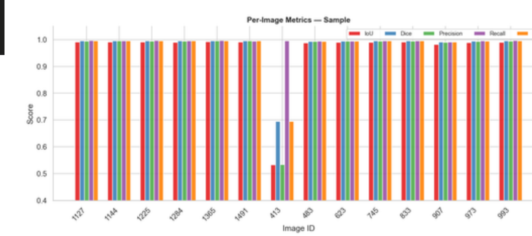
- Both Precision and Recall about 0.90 on a 5% positive call-the threshold for industrial usability

Class-aware loss is essential

- Without 4x pore oversampling and pos_weight=10, the pore head collasped to predicting empty masks

Methodolgy over metrics

- Identical failure on image 413 across two unrelated points to a data issue, not a model.



Future Work

- Train more models for training instead of just one
- Implement more images with inaccurate finding
- Fix the wraping from polar coordinates to cartesian coordinates

References

Woo, S., Park, J., Lee, J.-Y., & Kweon, I. S. (2018). CBAM: Convolutional block attention module. In Proceedings of the European Conference on Computer Vision (ECCV).

He, K., et al. (2016). Deep Residual Learning for Image Recognition. CVPR.

Lin, T.-Y., et al. (2017). Feature Pyramid Networks for Object Detection. CVPR.